Practice workbook · Identify alimentary canal organs and describe what each does

Naming, labelling or locating digestive organs from a diagram or feature, and describing the job of the mouth, pancreas, small or large intestine.

12 question parts: 1 worked-example retry and 11 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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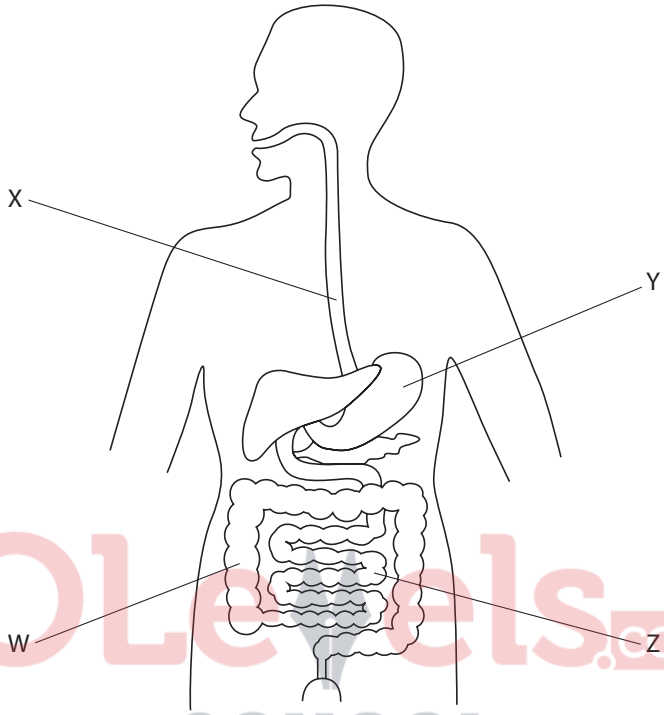
Type ID: HBI-GUT-ORGANS-IDENTIFY-AND-ROLE

1. Question 2 · part a.i · 4 marks

June 2022 Paper 1R

Worked-example retry · Attempt part a.i only.

2 (a) The diagram shows the human digestive system.



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(i) Name the structures labelled W, X, Y and Z.

(4)

W .....

X .....

Y .....

Z .....

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1. Question 2 · part a.i · 4 marks

June 2022 Paper 1R

Worked-example retry · Attempt part a.i only.

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(ii) The box gives words used to describe how food is moved through structure X towards structure Y.

relax	coordination	contract	liquid
peristalsis	muscles	enzymes	bolus

Use words from the box to complete the passage.

(4)

Food forms a ..... at the back of the mouth.

The food is swallowed and passes into structure X. The walls of structure X contain .....

which ..... to push the food towards structure Y.

This process is known as ..... .

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1. Question 2 · part a.i · 4 marks · Working and answer

Working and answer

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2. Question 1 · part whole · 6 marks

November 2020 Paper 1

Further practice · Attempt part whole only.

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1 The table lists some features and regions of the human alimentary canal.

Tick (✓) one box in each row to show the region of the alimentary canal where each feature occurs.

(6)

Feature	Region of alimentary canal			
	mouth	**stomach**	**small intestine**	**large intestine**
starts protein digestion				
starts carbohydrate digestion				
has a pH of 2				
has villi				
absorbs most water				
connects with bile duct				

(Total for Question 1 = 6 marks)

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2. Question 1 · part whole · 6 marks · Working and answer

Working and answer

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3. Question 1 · part a · 3 marks

November 2020 Paper 2R

Further practice · Attempt part a only.

1 Food enters the alimentary canal through the mouth.

All undigested food leaves the body through the anus.

(a) Name three structures the food passes through between the mouth and the anus.

(3)

1 .....

2 .....

3 .....

(b) Some of the food is digested by enzymes as it passes through the alimentary canal.

(i) Name an enzyme that digests protein.

(1)

.....

(ii) Describe the processes that occur in the mouth to digest food.

(4)

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3. Question 1 · part a · 3 marks · Working and answer

Working and answer

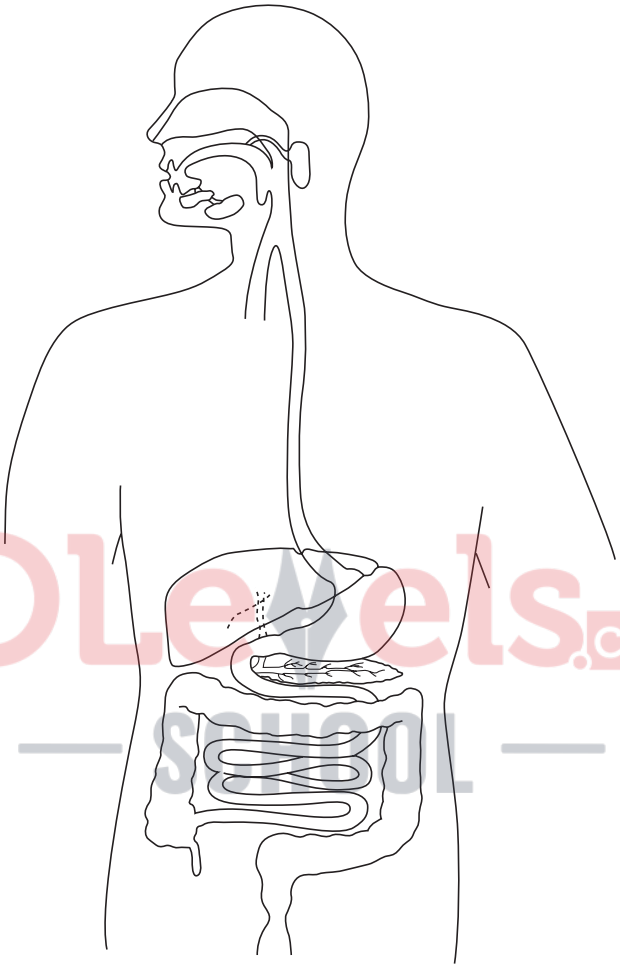
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4. Question 4 · part a.i · 1 mark

November 2021 Paper 1

Further practice · Attempt part a.i only.

4 The diagram shows the alimentary canal.



(a) Identify these areas of the alimentary canal using label lines and the correct letters.

(i) Label with an X the area where starch digestion begins.

(1)

(ii) Label with a Y the area that has a pH of 2

(1)

(iii) Label with a Z the area where both lipid and protein digesting enzymes are produced.

(1)

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P 6 7 0 6 0 R R A 0 8 2 0

4. Question 4 · part a.i · 1 mark · Working and answer

Working and answer

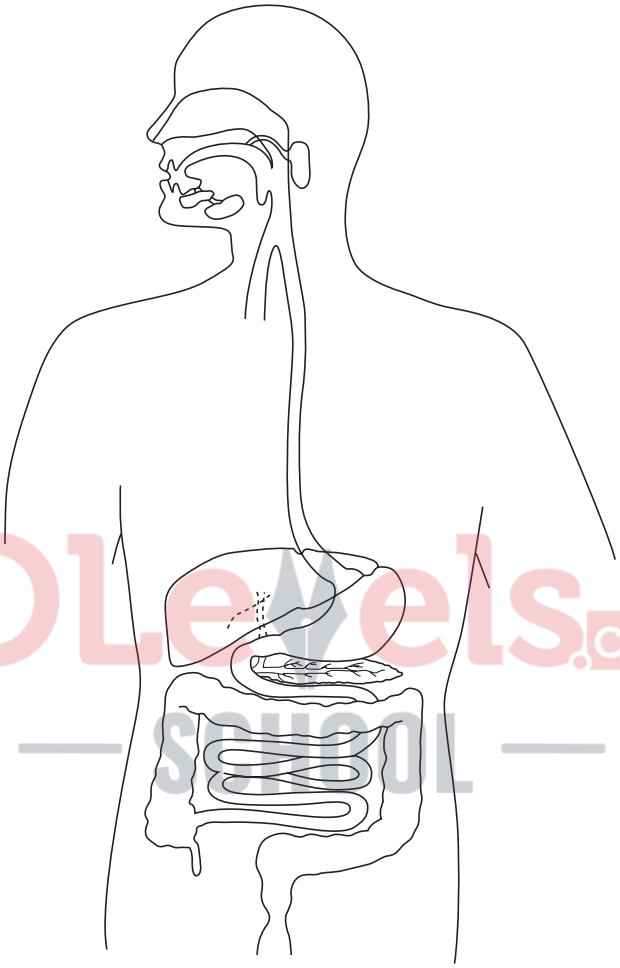
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5. Question 4 · part a.ii · 1 mark

November 2021 Paper 1

Further practice · Attempt part a.ii only.

4 The diagram shows the alimentary canal.



(a) Identify these areas of the alimentary canal using label lines and the correct letters.

(i) Label with an X the area where starch digestion begins.

(1)

(ii) Label with a Y the area that has a pH of 2

(1)

(iii) Label with a Z the area where both lipid and protein digesting enzymes are produced.

(1)

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5. Question 4 · part a.ii · 1 mark · Working and answer

Working and answer

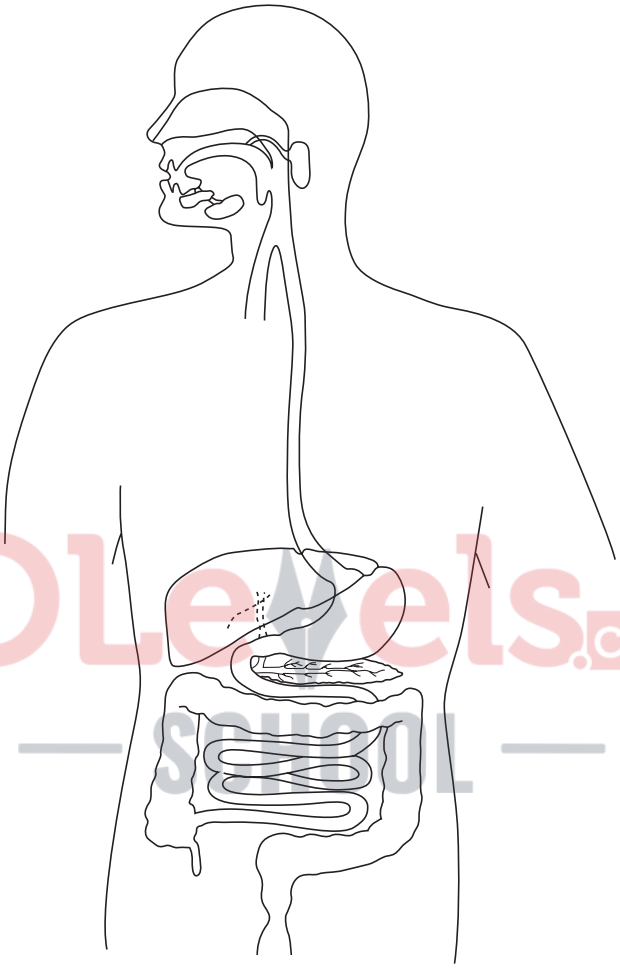
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6. Question 4 · part a.iii · 1 mark

November 2021 Paper 1

Further practice · Attempt part a.iii only.

4 The diagram shows the alimentary canal.



(a) Identify these areas of the alimentary canal using label lines and the correct letters.

(i) Label with an X the area where starch digestion begins.

(1)

(ii) Label with a Y the area that has a pH of 2

(1)

(iii) Label with a Z the area where both lipid and protein digesting enzymes are produced.

(1)

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6. Question 4 · part a.iii · 1 mark

November 2021 Paper 1

Further practice · Attempt part a.iii only.

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(b) Some people have the condition known as coeliac disease. This results in the villi being flattened.

Explain why some people with untreated coeliac disease will eventually develop osteoporosis.

(4)

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(c) Another condition of the alimentary canal is colon cancer. This is often treated by removal of the large intestine.

Explain why people who have this treatment often suffer from dehydration.

(2)

(Total for Question 4 = 9 marks)

6. Question 4 · part a.iii · 1 mark · Working and answer

Working and answer

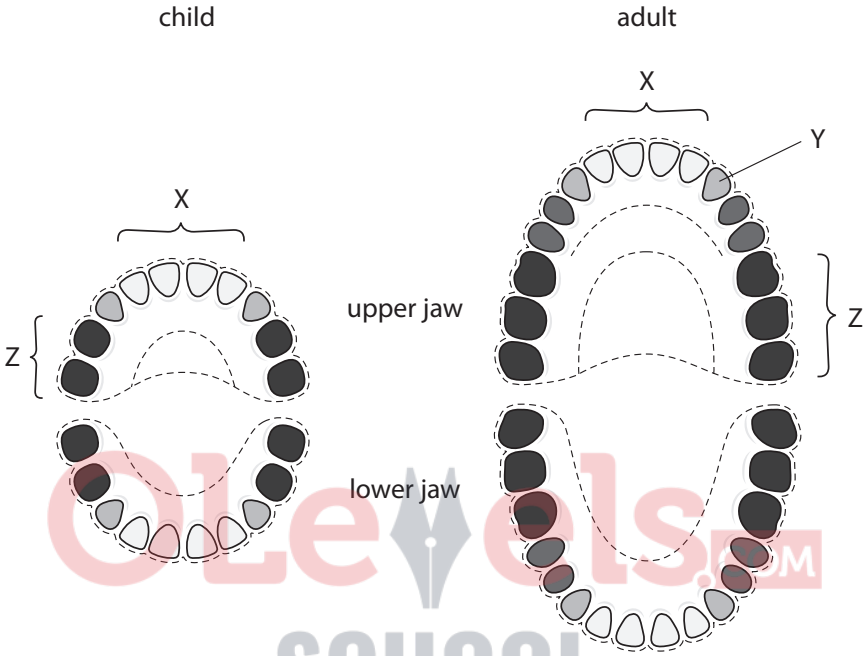
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7. Question 1 · part c.i · 1 mark

June 2024 Paper 2R

Further practice · Attempt part c.i only.

1 (a) The diagrams show the arrangement of teeth in a young child and the arrangement of teeth in an adult.



(Source: https://www.medicalnewstoday.com/articles/326754#number-of-teeth)

(i) Name the types of teeth shown by X and Z.

(2)

X .....

Z .....

(ii) State the function of the tooth labelled Y.

(1)

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P 7 4 4 9 5 A 0 2 2 8

7. Question 1 · part c.i · 1 mark

June 2024 Paper 2R

Further practice · Attempt part c.i only.

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(b) Digestion of food begins in the mouth.

Use the most appropriate words or numbers to complete the passage about digestion of food in the mouth.

(4)

Saliva in the mouth contains an enzyme called ..... .

This enzyme speeds up the digestion of starch to produce

..... .

The enzyme in saliva works best at pH ..... .

If the pH in the mouth was too low or too high, the digestion of carbohydrates

would ..... .

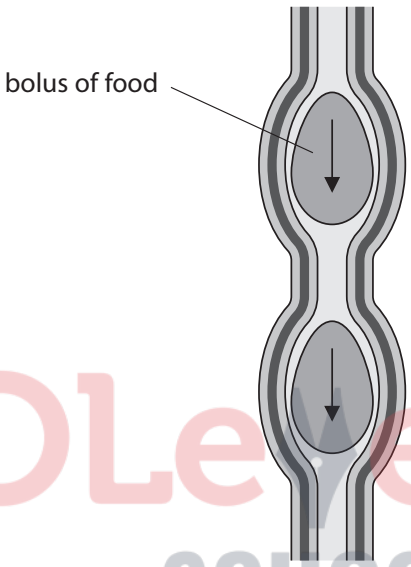
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7. Question 1 · part c.i · 1 mark

June 2024 Paper 2R

Further practice · Attempt part c.i only.

(c) The diagram shows the part of the digestive system that transports food from the mouth to the stomach.



(Source: https://www.motifolio.com/1031265.html)

(i) Name the part of the digestive system that transports food from the mouth to the stomach.

(1)

.....

(ii) Describe how the action of this part of the digestive system transports food from the mouth to the stomach.

(2)

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(Total for Question 1 = 10 marks)

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7. Question 1 · part c.i · 1 mark · Working and answer

Working and answer

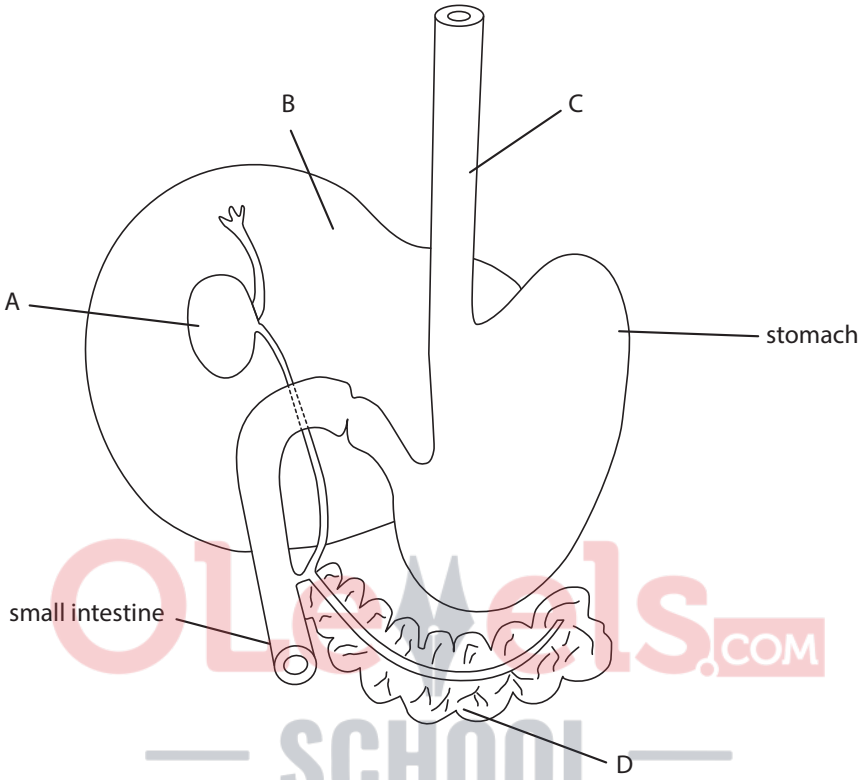
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8. Question 2 · part a.i · 4 marks

June 2025 Paper 2

Further practice · Attempt part a.i only.

2 The diagram shows structures involved in digestion.



(a) (i) Identify the structures labelled A, B, C and D.

(4)

A .....

B .....

C .....

D .....

(ii) Name the fluid stored in structure A.

(1)

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8. Question 2 · part a.i · 4 marks · Working and answer

Working and answer

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9. Question 4 · part b · 2 marks

January 2020 Paper 2

Further practice · Attempt part b only.

4 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Pinworms are small parasites that live in the small intestine and colon of humans. They emerge at night and lay their eggs in the anal region. In addition to laying over 10 000 eggs at one time, pinworms also secrete a substance that causes a very strong itching sensation around the anus. Although this is particularly uncomfortable for the host, itching plays an important role in the spread of infection.

Pinworm infections are highly contagious and occur most commonly in children under the age of five years. Infections occur when eggs are ingested or inhaled. These eggs have been deposited onto a surface or object by an infected person. The eggs, which are very tiny, can survive on these surfaces for several weeks.

One method of detecting a pinworm infection is to carry out a tape test. This test consists of sticking a piece of transparent tape to the anus in the morning. If an infection is detected then medication can be taken to treat it. In addition to taking medication, it is important that the infected person and other people in the house follow strict procedures to eliminate pinworm eggs and to prevent the spread of the infection.

In rare cases where infections are not treated, large populations of pinworms can grow in the intestines. This can lead to nutritional deficiencies in the infected person, which cause further symptoms that have a more serious effect on health.

(a) A pinworm infection causes itching.

(i) Give one other symptom of a pinworm infection.

(1)

(ii) Explain why itching plays an important role in the spread of a pinworm infection (lines 4 to 5).

(2)

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9. Question 4 · part b · 2 marks

January 2020 Paper 2

Further practice · Attempt part b only.

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(b) Suggest how the small intestine provides perfect conditions for the growth of pinworm populations.

(2)

(c) Pinworms can cause nutritional deficiencies (line 16).

(i) State what is meant by the term **nutritional deficiency**.

(1)

(ii) State one nutritional deficiency that could be caused by a pinworm infection.

(1)

(iii) Suggest why a large population of pinworms in a person's intestines can lead to nutritional deficiency.

(2)

(d) Suggest how the transparent tape test can be used to detect a pinworm infection (lines 10 to 11).

(1)

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9. Question 4 · part b · 2 marks · Working and answer

Working and answer

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10. Question 1 · part b.ii · 4 marks

November 2020 Paper 2R

Further practice · Attempt part b.ii only.

1 Food enters the alimentary canal through the mouth.

All undigested food leaves the body through the anus.

(a) Name three structures the food passes through between the mouth and the anus.

(3)

1 .....

2 .....

3 .....

(b) Some of the food is digested by enzymes as it passes through the alimentary canal.

(i) Name an enzyme that digests protein.

(1)

.....

(ii) Describe the processes that occur in the mouth to digest food.

(4)

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10. Question 1 · part b.ii · 4 marks

November 2020 Paper 2R

Further practice · Attempt part b.ii only.

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(c) Fibre in food cannot be digested.

Explain the importance of fibre in the diet.

(3)

(Total for Question 1 = 11 marks)

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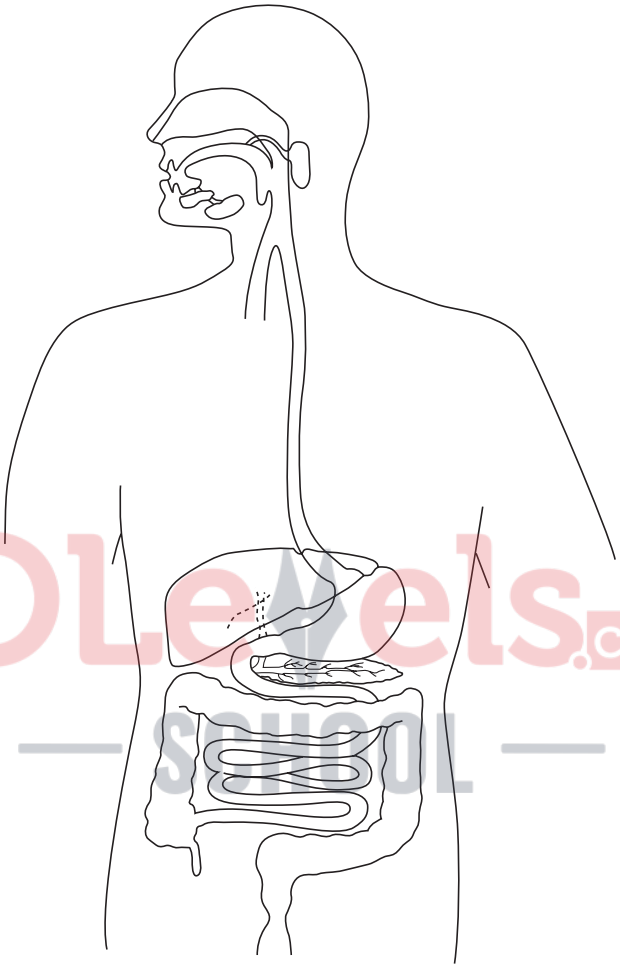
Working and answer

11. Question 4 · part c · 2 marks

November 2021 Paper 1

Further practice · Attempt part c only.

4 The diagram shows the alimentary canal.



(a) Identify these areas of the alimentary canal using label lines and the correct letters.

(i) Label with an X the area where starch digestion begins.

(1)

(ii) Label with a Y the area that has a pH of 2

(1)

(iii) Label with a Z the area where both lipid and protein digesting enzymes are produced.

(1)

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11. Question 4 · part c · 2 marks

November 2021 Paper 1

Further practice · Attempt part c only.

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(b) Some people have the condition known as coeliac disease. This results in the villi being flattened.

Explain why some people with untreated coeliac disease will eventually develop osteoporosis.

(4)

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(c) Another condition of the alimentary canal is colon cancer. This is often treated by removal of the large intestine.

Explain why people who have this treatment often suffer from dehydration.

(2)

(Total for Question 4 = 9 marks)

11. Question 4 · part c · 2 marks · Working and answer

Working and answer

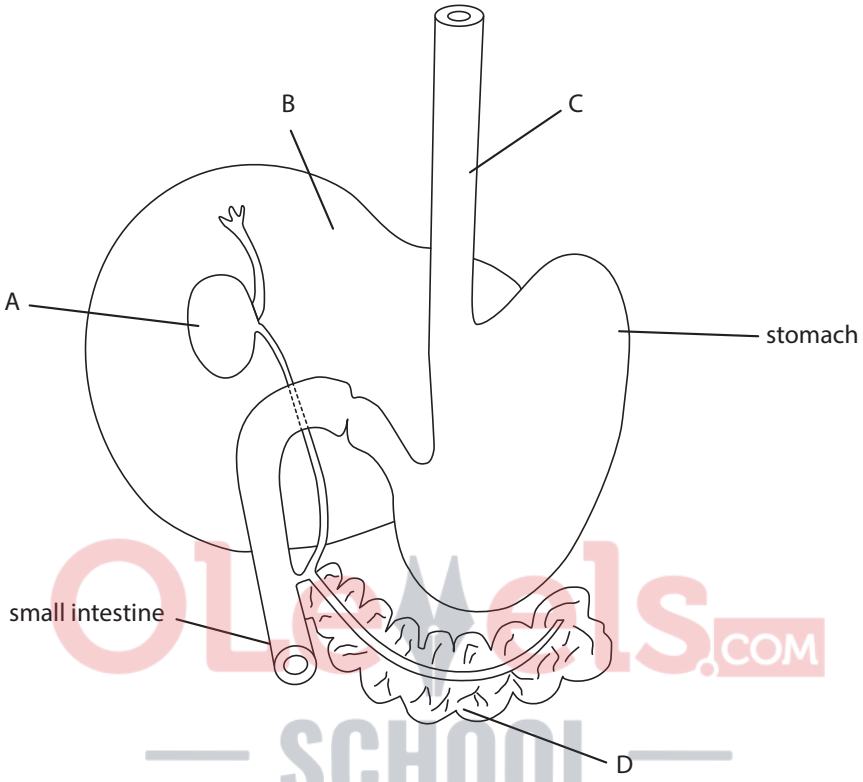
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12. Question 2 · part b.i · 4 marks

June 2025 Paper 2

Further practice · Attempt part b.i only.

2 The diagram shows structures involved in digestion.



(a) (i) Identify the structures labelled A, B, C and D.

(4)

A .....

B .....

C .....

D .....

(ii) Name the fluid stored in structure A.

(1)

.....

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12. Question 2 · part b.i · 4 marks

June 2025 Paper 2

Further practice · Attempt part b.i only.

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(iii) Explain the function of the fluid stored in structure A in digestion.

(3)

(b) (i) Describe the role of structure D in digestion.

(4)

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(ii) Name two substances secreted by structure D that are not involved in digestion.

(2)

1 .....

2 .....

(Total for Question 2 = 14 marks)

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12. Question 2 · part b.i · 4 marks · Working and answer

Working and answer

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